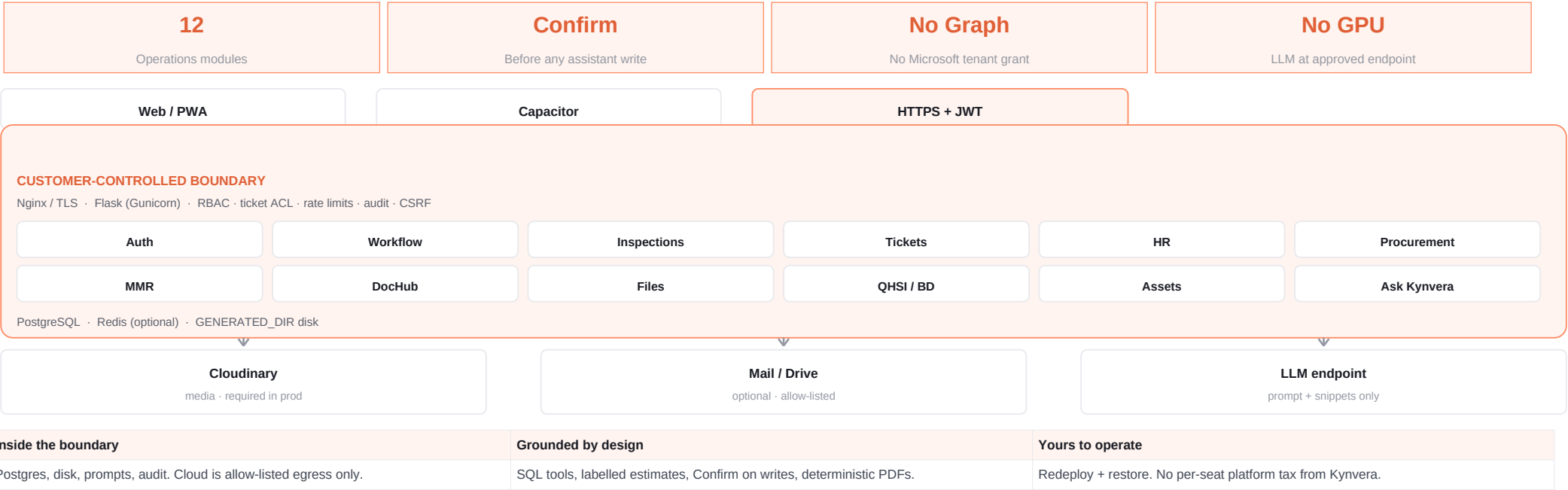


SOLUTION ARCHITECTURE

# Kynvera — Injaaz FM Operations Platform

Inside your operations stack · Grounded & auditable · Yours to operate · All Operations. One Platform.

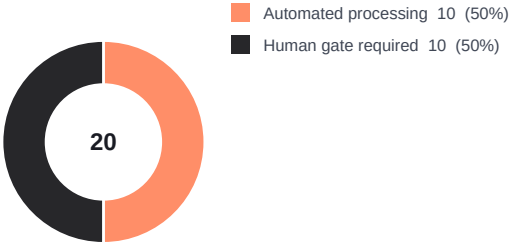


THE OPERATING TEAM

Modules own stages of FM work. Ask Kynvera sits across them.

|                                                                                                                    |                                                                                                                      |
|--------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------|
| <div>Ask Kynvera</div> <div>Auto Grounded Q&amp;A; draft proposals</div> <div>Human Confirm before any write</div> | <div>Inspections</div> <div>Auto HVAC / Civil / Cleaning, PDF</div> <div>Human Reviewer signatures</div>             |
| <div>Ticketing</div> <div>Auto Lifecycle, photos, SLA, costs</div> <div>Human Assign, verify, close, markup</div>  | <div>Email intake</div> <div>Auto Mail → draft ticket</div> <div>Human Convert draft to live work</div>              |
| <div>AI triage</div> <div>Auto Priority / SLA / technician / parts</div> <div>Human Accept or override</div>       | <div>Assets / GIS</div> <div>Auto Registry, QR, map, 2D twin</div> <div>Human Master data decisions</div>            |
| <div>HR</div> <div>Auto Leave and workforce forms</div> <div>Human HR → GM chain</div>                             | <div>Procurement</div> <div>Auto Catalogues, PRs</div> <div>Human Threshold / GM rules</div>                         |
| <div>MMR</div> <div>Auto CAFM ingest, chargeable rules</div> <div>Human Recipients you configure</div>             | <div>DocHub / Files</div> <div>Auto Controlled docs; optional Drive</div> <div>Human Publish and folder policy</div> |

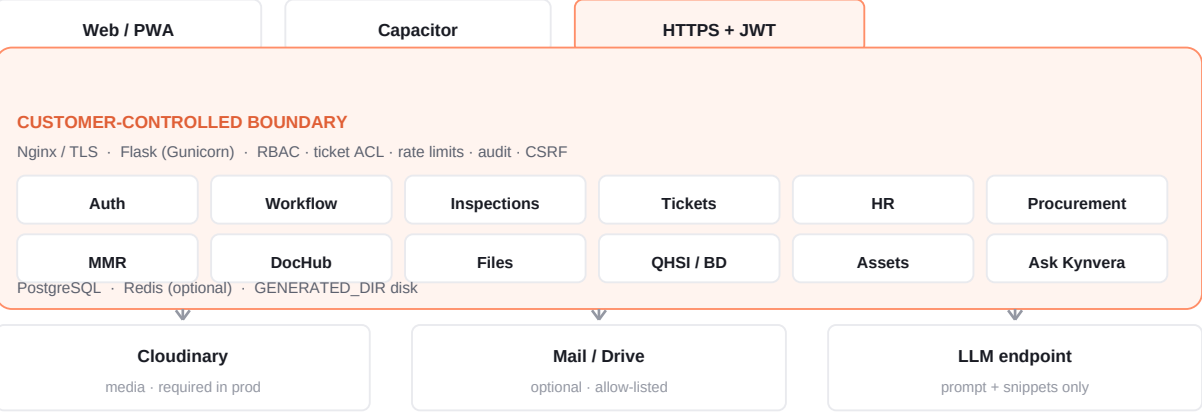
Stages: process vs sign-off



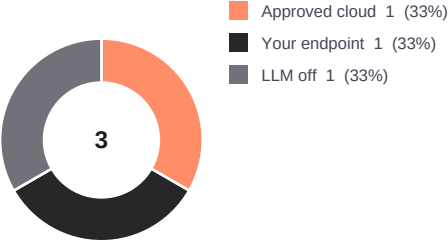
Source: module count in the product. Automation handles processing and drafts; people approve, assign, sign, and close. Nothing client-facing goes out on assistant say-so.

DATA SOVEREIGNTY

Records stay. Inference is a single logged path you approve.



LLM modes (options, not usage)



Relative exposure classes

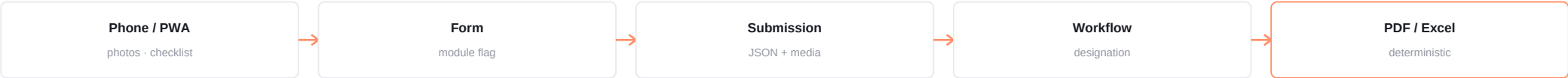


| Class                      | Where it lives                      | May leave?                          |
|----------------------------|-------------------------------------|-------------------------------------|
| Tickets, forms, HR, assets | PostgreSQL                          | No — never as a bulk dump           |
| Knowledge / FAQs / DocHub  | Postgres + DocHub ACL               | Only retrieved excerpts in a prompt |
| Photos / signatures        | Cloudinary (prod) or disk           | Yes — contracted media account      |
| Generated PDF / Excel      | GENERATED_DIR                       | No unless you email them            |
| Assistant writes           | Pending action 15 min, then Confirm | No unattended submit                |

Cloud LLM: current turn only. Provider API terms do not train on API traffic by default, subject to your agreement.

DATA FLOW A — FIELD CAPTURE

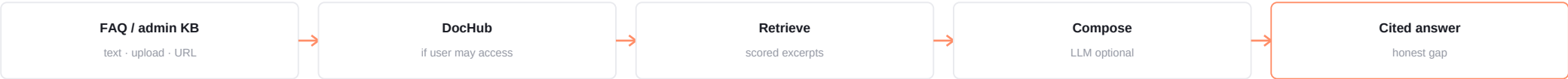
Site visit to signed, auditable record



Media to Cloudinary or disk — not to the LLM unless a later permitted question retrieves it.

DATA FLOW B — KNOWLEDGE

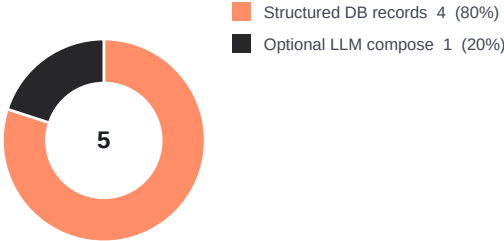
Help content becomes answers without dumping the database



Ingestion stays in Postgres/DocHub. Only compose may call a model.

| Form class      | Stored as                     | Completeness            |
|-----------------|-------------------------------|-------------------------|
| Inspection / HR | Submission + JSON form_data   | Workflow + signatures   |
| Work orders     | Ticket rows + images + SLA    | Draft vs live separated |
| Assets          | Asset registry + QR + lat/lng | CRUD + ticket link      |
| CAFM / MMR      | Uploaded workbooks + rules    | Chargeable checks       |

Capture vs compose



DATA FLOW C — ASK KYNVERA

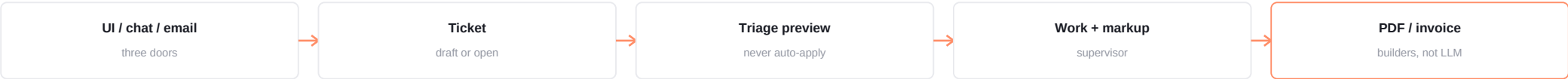
A question becomes a cited answer — or a Confirm card



Restricted tools return allowed:false. Restricted material never enters the prompt.

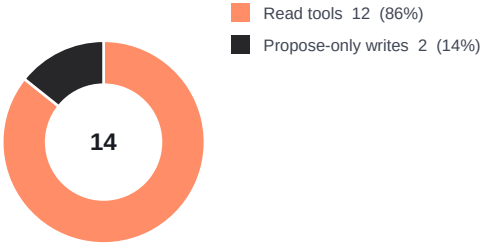
DATA FLOW D — WORK ORDERS

Three intakes. One ticket. Humans own commercial numbers.



Markup % and chargeable flags are supervisor fields — the assistant has no cost-estimator write tool.

Ask Kynvera tools (count)

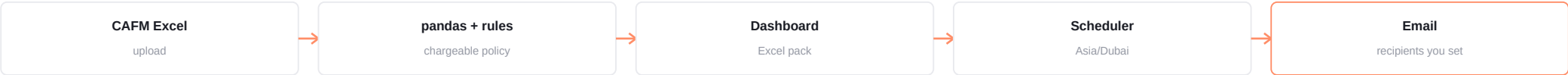


Source: module\_assistant tool registry — 12 read tools, 2 propose tools (ticket draft, leave draft).

RAG steps inside vs egress

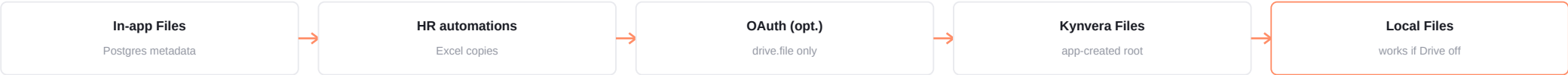


DATA FLOW E — MMR / CAFM



Workbook stays in app/disk. Email is an approved outbound path.

DATA FLOW F — FILES AND DRIVE



drive.file cannot read the rest of the user's Drive.

| Control            | Meaning                                                                 |
|--------------------|-------------------------------------------------------------------------|
| Source-bound       | Ticket/leave/FM numbers from SQL; policies from knowledge/DocHub        |
| Gaps flagged       | No source → say so; triage returns null technician if none fit          |
| Estimates labelled | Asset RUL stored as method=llm_estimate until a trained model exists    |
| Human-in-the-loop  | Confirm, workflow sign, email-draft conversion                          |
| Deterministic docs | Client PDFs, invoices, MMR packs are ReportLab/openpyxl — not LLM prose |

WHERE IT RUNS · HARDENING · ADOPTION

Right-sized VM. GPU unused. Degrades gracefully. VAPT is your production gate if required.

| Component   | Reference            | Notes                    | Failure         | Behaviour                           |
|-------------|----------------------|--------------------------|-----------------|-------------------------------------|
| Application | 2–4 vCPU, 8–16 GB    | Gunicorn; GPU unused     | LLM down        | FAQ/intent fallback; forms continue |
| PostgreSQL  | Managed              | Required in production   | Cloudinary down | New uploads fail; DB intact         |
| Redis       | 256 MB+              | Rate limits; optional RQ | Mail down       | UI continues; mail delayed          |
| Disk        | 100 GB+ SSD          | GENERATED_DIR reports    | Drive down      | Local Files continues               |
| OS          | Ubuntu 22.04 / 24.04 | Docker user injaaz       | Redis down      | App may run; limits degraded        |
|             |                      |                          | App VM fails    | Redeploy + restore Postgres + disk  |

| Layer                  | What exists today                                          |
|------------------------|------------------------------------------------------------|
| Build                  | Pinned deps, secrets in env, non-root container            |
| Identity               | bcrypt, JWT + session JTI blocklist, rate-limited login    |
| Transport / headers    | TLS, HSTS, nosniff, SAMEORIGIN, CSP Report-Only            |
| AuthZ / AI             | Module flags, ticket ACL, tool ACL, 15-min pending actions |
| Not claimed as default | Third-party pen-test, ISO/SOC, Entra SSO, TOTP MFA         |



Each phase is useful alone. No rip-and-replace of ERP. Next: security review → confirm LLM, Cloudinary, mail, Drive → contained pilot.